Extended Essay

π

Subject: Mathematics Analysis and Approaches (HL)

Topic: Methods to Calculate π

Research Question:

"Which amongst John Wallis Product Formula, Archimedes Method, Francois Viète Formula, Newton's Formula and Gregory-Leibniz Series is the most efficient method to approximate the value of π?"

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INTRODCUTION

π has been known for approximately 4000 years, yet even if we counted the seconds in that time and approximated the value of π by the number of decimal places, we would still just be approximating the true value of π. π is an irrational number which means that it can't be expressed as a ratio of 2 numbers. The fact that it's a non-recurring and non-terminating decimal, it does not settle into an eternally repeating pattern of digits, despite the fact that fractions like $\frac{22}{7}$ and $\frac{355}{133}$ are frequently used to approximate it, means that these cannot be used to represent its true value. By multiplying the circle's radius by three, the ancient Babylonians determined its area, yielding the result $\pi = 3$. Finally, they reached a conclusion and discovered that $\pi = 3.125$, a closer approximation. [1]

Worthy of Investigation

Numerous mathematical formulae have made use of the sign (π) throughout the history of mathematics' progression as a discipline. It is used in a number of different computations that are necessary for the functioning of the contemporary world. It is necessary for us to do an analysis of π since there are a great number of spherical objects in our universe. The constant, π, is used in every calculation that is carried out by structural engineers who are entrusted with the construction of structures that are able to withstand earthquakes. In addition, the value of π offers support to architects working in the area of architecture throughout the process of designing vehicles and equipment that is, among other things, necessary for the process of building landmarks and establishing new ones. Using the symbol to make an approximation of the value of an infinite series is often possible, at least to some degree. As a result of this, mathematicians from all over the globe may take use of it since it makes it easier for them to carry out the necessary calculations. For the sake of space exploration, for example, astronauts from NASA use the value of π to deploy spacecrafts into orbit around other planets in order to gather data[2]. This is something that is necessary and helpful in the

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[1] "A Brief History Of Pi (Π) | Exploratorium". *Exploratorium*, 2017, https://www.exploratorium.edu/pi/history-of-pi#:~:text=Mathematicians%20began%20using%20the%20Greek,calculate%20%CF%80%20based%20on%20probability. Accessed 6 Feb 2023.

[2] Atkinson, Nancy. "Happy Pi Day: 5 Ways NASA Uses Pi". *Universe Today*, 2014, https://www.universetoday.com/110331/happy-pi-day-5-ways-nasa-uses-pi/#:~:text=In%20basic%20mathematics%2C%20Pi%20is,flight%2C%20to%20name%20a%20few. Accessed 6 Feb 2023.

modern world because of how much it is spreading. Tacticians use π to keep track of population patterns, while biochemists utilize it when trying to comprehend the structure of DNA[3]. These are two more examples of notable applications of π in the real world.

Research Question:

Which amongst John Wallis Product Formula, Archimedes Method, Francois Viète Formula, Newton's Formula and Gregory-Leibniz Series is the most efficient method to approximate the value of π?

Research Methodology-

For the aim of this Extended Essay, I will describe how to calculate π using several approaches, such as Archimedes method, Isaac Newton's Series, Gregory-Leibniz Series, John Wallis Product Formula, and Francois Viète's Series. After that, I will determine the reliability of these procedures by contrasting the results with a value that is already established. Because these methods are the most well-known and dependable ones that I could find, I've opted to use them. I attempted to compute to the accuracy of π up to 14th decimal place by using Microsoft Excel. This is the most precise value that the spreadsheet is capable of producing. This was carried out in a separate fashion for each approach. I applied the algorithm to three hundred distinct phrases in order to determine which method is the most efficient, and then I estimated the accuracy in order to select the strategy that was the most successful overall. I have chosen to approximate the value of π up till 14 decimal places as the astronauts from NASA deem it necessary for the value to be at least 14 decimal places. The method that would approximate the most correct value while using the fewest iterations feasible would be considered the most efficient way to approximate the value of π.

[3] Wilson, Katie A. et al. "DNA–Protein Π-Interactions In Nature: Abundance, Structure, Composition And Strength Of Contacts Between Aromatic Amino Acids And DNA Nucleobases Or Deoxyribose Sugar". *Nucleic Acids Research*, vol 42, no. 10, 2014, pp. 6726-6741. *Oxford University Press (OUP)*, doi:10.1093/nar/gku269. Accessed 14 Feb 2023.

Francois Viète's Formula for approximating π-

Francois Viète, a French mathematician who lived from 1540 to 1603, was the first person in recorded history to write a formula that expressed a limit. This formula was published in 1593. Using this algorithm, he was capable of doing calculations with an accuracy of nine digits[4].

Proof-

We would need to demonstrate two trigonometric identities before deriving,

$$\sin\theta = 2\sin\theta\cos\theta$$

And

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\cos(\theta)}{2}}$$

Using Euler's formula, I will prove this,

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Formula when angle is equal to $\frac{\theta}{2}$,

$$e^{i\frac{\theta}{2}} = \cos\frac{\theta}{2} + i\sin\frac{\theta}{2}$$

This can also be written as,

$$(e^{i\theta})^{\frac{1}{2}} = (\cos\theta + i\sin\theta)^{\frac{1}{2}}$$

$$\therefore \cos\frac{\theta}{2} + i\sin\frac{\theta}{2} = (\cos\theta + i\sin\theta)^{\frac{1}{2}}$$

Squaring the whole equation,

$$\left(\cos\frac{\theta}{2}\right)^2 + 2i\cos\frac{\theta}{2}\sin\frac{\theta}{2} + i^2\left(\sin\frac{\theta}{2}\right)^2 = \cos\theta + i\sin\theta$$

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[4] Published byPRESH TALWALKARI run the MindYourDecisions channel on YouTube *et al.* (no date) *The Wallis product formula for PI and its proof*, *Mind Your Decisions*. Available at: https://mindyourdecisions.com/blog/2016/10/12/the-wallis-product-formula-for-pi-and-its-proof/#:~:text=Start%20with%20the%20fractions%202%2F1%20and%202%2F3.&text=Each%20new%20pair%20is%20%5B2n,to%20%CF%80%2F2%20%E2%89%88%201.571. (Accessed: February 8, 2023).

$$\left(\cos\frac{\theta}{2}\right)^2 - \left(\sin\frac{\theta}{2}\right)^2 + 2i\cos\frac{\theta}{2}\sin\frac{\theta}{2} = \cos\theta + i\sin\theta$$

Equating similar values,

$$\cos\theta = \left(\cos\frac{\theta}{2}\right)^2 - \left(\sin\frac{\theta}{2}\right)^2$$

$$\sin\theta = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}$$

We now have our first identity, and we can use the Pythagorean identity for our second identity, $\left(\sin\frac{\theta}{2}\right)^2 = 1 - \left(\cos\frac{\theta}{2}\right)^2$ to get,

$$\cos\theta = \left(\cos\frac{\theta}{2}\right)^2 - \left(1 - \left(\cos\frac{\theta}{2}\right)^2\right)$$

$$\cos\theta = 2\left(\cos\frac{\theta}{2}\right)^2 - 1$$

$$\frac{\cos\theta + 1}{2} = \left(\cos\frac{\theta}{2}\right)^2$$

$$\therefore \cos\frac{\theta}{2} = \sqrt{\frac{\cos\theta + 1}{2}}$$

We can derive Viète's Series by sing both of these identities.

Firstly, we'll use the identity we found,

$$\sin\theta = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}$$

We can prove by using this identity,

$$\sin\frac{\theta}{2} = 2\cos\frac{\theta}{4}\sin\frac{\theta}{4}$$

$$\sin\frac{\theta}{4} = 2\cos\frac{\theta}{8}\sin\frac{\theta}{8}$$

Therefore,

$$\sin\theta = 2\cos\frac{\theta}{2}\left(2\cos\frac{\theta}{4}\sin\frac{\theta}{4}\right)$$

$$\sin\theta = 2\cos\frac{\theta}{2}\left(2\cos\frac{\theta}{4}\left(2\cos\frac{\theta}{8}\sin\frac{\theta}{8}\right)\right)$$

This can be generalized as,

$\sin\theta = 2^k \times \cos\frac{\theta}{2} \times \cos\frac{\theta}{4} \times \cos\frac{\theta}{8} \times \ldots \cos\frac{\theta}{2^k} \times \sin\frac{\theta}{2^k}$ where 'k' is a real integer.

Dividing the whole equation by θ,

$$\frac{\sin\theta}{\theta} = \frac{2^k}{\theta} \times \cos\frac{\theta}{2} \times \cos\frac{\theta}{4} \times \cos\frac{\theta}{8} \times \ldots \cos\frac{\theta}{2^k} \times \sin\frac{\theta}{2^k}$$

This can be written as,

$$\frac{\sin\theta}{\theta} = \cos\frac{\theta}{2} \times \cos\frac{\theta}{4} \times \cos\frac{\theta}{8} \times \ldots \cos\frac{\theta}{2^k} \times \frac{\sin\frac{\theta}{2^k}}{\frac{\theta}{2^k}}$$

$$\lim_{n\to\infty} \frac{\sin\frac{\theta}{2^k}}{\frac{\theta}{2^k}} = 1$$

As $k \to \infty$, $\sin\frac{\theta}{2^k} \to \sin(0)$ and $\frac{\theta}{2^k} \to 0$

Therefore, we can also write it as-

$$\lim_{x\to 0} \frac{\sin(x)}{x} = 1$$

$$\therefore \frac{\sin\theta}{\theta} = \cos\frac{\theta}{2} \times \cos\frac{\theta}{4} \times \cos\frac{\theta}{8} \times \ldots$$

When $\theta = \frac{\pi}{2}$,

$$\frac{1}{\frac{\pi}{2}} = \cos\frac{\pi}{4} \times \cos\frac{\pi}{8} \times \cos\frac{\pi}{16} \times \ldots$$

Using the second identity $\cos\frac{\theta}{2} = \sqrt{\frac{\cos\theta+1}{2}}$ we can simplify it further-

As $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$

$$\cos\left(\frac{\pi}{8}\right) = \sqrt{\frac{1+\cos\frac{\pi}{4}}{2}}$$

$$\cos\left(\frac{\pi}{8}\right) = \sqrt{\frac{1+\frac{\sqrt{2}}{2}}{2}}$$

$$\cos\left(\frac{\pi}{8}\right) = \sqrt{\frac{\sqrt{2}+2}{4}}$$

$$\cos\left(\frac{\pi}{8}\right) = \frac{\sqrt{\sqrt{2}+2}}{2}$$

Similarly,

$$\cos\left(\frac{\pi}{16}\right) = \left(\frac{1+cos\frac{\pi}{8}}{2}\right)^{\frac{1}{2}}$$

$$\cos\left(\frac{\pi}{16}\right) = \left(\frac{1+\frac{\sqrt{2+\sqrt{2}}}{2}}{2}\right)^{\frac{1}{2}}$$

$$\cos\left(\frac{\pi}{16}\right) = \left(\frac{2+\sqrt{2+\sqrt{2}}}{4}\right)^{\frac{1}{2}}$$

$$\cos\left(\frac{\pi}{16}\right) = \frac{\left(\sqrt{2+\sqrt{2}}+2\right)^{\frac{1}{2}}}{2}$$

Eventually, we would get-

$$\frac{2}{\pi} = \frac{\sqrt{2}}{2} \times \frac{\sqrt{2+\sqrt{2}}}{2} \times \frac{\sqrt{2+\sqrt{2+\sqrt{2}}}}{2} \times \dots$$

Therefore,

$$\pi = 2 \times \frac{2}{\sqrt{2}} \times \frac{2}{\sqrt{2+\sqrt{2}}} \times \frac{2}{\sqrt{2+\sqrt{2+\sqrt{2}}}} \times \dots$$

Following the completion of this series to create a spreadsheet, the anticipated value of π after only the 24th term of this infinite product, we obtain π to the 14th decimal place. This value is an exact match to the real value of π till 14 decimal places. This product series is the first to correctly calculate the value of π and only required the multiplication of the first 24 terms, demonstrating its accuracy and speed.

Table 1:

Approximation of π in a table using Francois Viète's Formula till 24 terms[5]:

No. of terms	n-sided polygon	Approximation of pi
1	4	2.82842712474619
2	8	3.06146745892072
3	16	3.12144515225805
4	32	3.13654849054594
5	64	3.14033115695475
6	128	3.14127725093277
7	256	3.14151380114430
8	512	3.14157294036709
9	1024	3.14158772527716
10	2048	3.14159142151120
11	4096	3.14159234557012
12	8192	3.14159257658487
13	16384	3.14159263433856
14	32768	3.14159264877699
15	65536	3.14159265238659
16	131072	3.14159265328899
17	262144	3.14159265351459
18	524288	3.14159265357099
…	…	…
24	33554432	3.14159265358979

[5] Refer to Appendix, Francois Viète series data

John Wallis Product Formula-

It is stated that John Wallis largely relied on guesswork and intuition to come up with his series[6]. Since the sign for was not widely used at the time, Wallis created his own symbol to represent $\frac{4}{\pi}$: a square with a dot within. Wallis identified a pattern with integers and believed the same pattern would hold true with fractions.

Let's consider the class of integrals Wallis is trying to evaluate[7]:

$$\int_0^1 (1-x^2)^{\frac{n}{2}} dx$$

Let's substitute $x = \cos\theta$, causing the integration limits to range from $0\ to \frac{\pi}{2}$, and the formula, $\sin^2\theta = 1 - \cos^2\theta$, simplifies the integrand. Ultimately, we focus on the set of sines to the power of n integrals. It will be simpler to calculate if we alter the upper limit to π.

$$\int_0^{\pi} \sin^n\theta d\theta = I(n), \quad n = 0,1,2,3 \ldots$$

We can derive the Wallis product formula from these integrals. First, we will integrate by parts:

$$I(n) = \int_0^{\pi} \sin^n\theta d\theta$$

$$u = \sin^{n-1}\theta$$

$$du = (n-1)\sin^{n-2}\theta\cos\theta\, d\theta$$

$$dv = \sin\theta d\theta$$

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[6] *Wallis's product* (no date) *ProofWiki*. Available at: https://proofwiki.org/wiki/Wallis%27s_Product#:~:text=He%20did%20this%20job%20by,n%E2%88%922xdx (Accessed: February 14, 2023).

[7] Published byPRESH TALWALKARI run the MindYourDecisions channel on YouTube *et al.* (no date) *The Wallis product formula for PI and its proof*, *Mind Your Decisions*. Available at: https://mindyourdecisions.com/blog/2016/10/12/the-wallis-product-formula-for-pi-and-its-proof/#:~:text=Start%20with%20the%20fractions%202%2F1%20and%202%2F3.&text=Each%20new%20pair%20is%20%5B2n,to%20%CF%80%2F2%20%E2%89%88%201.571. (Accessed: February 8, 2023).

$v = -\cos\theta$

$$I(n) = \int_0^{\pi} \sin^n \theta \, d\theta$$

$$= \left[\sin^{n-1}\theta . -\cos\theta\right]_0^{\pi} - \int_0^{\pi} -\cos\theta \, du$$

$$= 0 - \int_0^{\pi} -\cos\theta (n-1)\sin^{n-2}\theta \cos\theta \, d\theta$$

$$= (n-1)\int_0^{\pi} (\cos^2\theta)\sin^{n-2}\theta \cos\theta \, d\theta$$

$$= (n-1)\int_0^{\pi} (1-\sin^2\theta)\sin^{n-2}\theta$$

$$= (n-1)\int_0^{\pi} \sin^{n-2}\theta \, d\theta - (n-1)\int_0^{\pi} \sin^n\theta \, d\theta$$

$$= [(n-1)\, I(n-2)] - [(n-1)\, I(n)]$$

$$\rightarrow I(n) = \frac{n-1}{n}\, I(n-2)$$

This recursive formula converts the integral with power n into an integral with power $n-2$.

We can write formulas for the ratios of integrals for even and odd integers of n.

$\frac{I(2n)}{I(2n-1)} = \frac{2n-1}{2n}$, where $'n'$ is an even integer.

$\frac{I(2n+1)}{I(2n-1)} = \frac{2n}{2n+1}$, where $'n'$ is an odd integer.

We have a recursive formula for the integral, so let us calculate a few of the smaller powers.

$$I(0) = \int_0^{\pi} d\theta = \pi$$

$$I(1) = \int_0^{\pi} \sin\theta \, d\theta = 2$$

Now we can use the recursive formula for even numbers:

$$I(2n) = \int_0^{\pi} \sin^{2n}\theta \, d\theta = \frac{2n-1}{2n} \, I(2n-2)$$

Similarly, for odd integers, we apply the recursive formula to I_{2n+1}, and then repeat until we get to $I_1 = 2$:

$$I(2n+1) = \int_0^{\pi} \sin^{2n+1}\theta \, d\theta$$

$$= \frac{2n}{2n+1} \, I(2n-1)$$

$$= \frac{2n}{2n+1}\frac{2n-2}{2n-1} \, I(2n-3)$$

$$= \frac{2n}{2n+1}\frac{2n-2}{2n-1} \ldots I(1)$$

$$= \frac{2n}{2n+1}\frac{2n-2}{2n-1} \ldots 2$$

$$= 2\prod_{k=1}^{n}\frac{2k}{2k+1}$$

Therefore, in this interval $0 \leq \theta \leq \pi$ we can say that $0 \leq \sin\theta \leq 1$. Increasing powers of sine in this interval form a non-increasing sequence, and that implies the integral values are also a non-increasing sequence $I(2n+1) \leq I(2n) \leq I(2n-1)$. Consequently, divide every term by $I(2n+1)$ in this inequality, and use $\frac{I(2n-1)}{I(2n+1)} = \frac{2n+1}{2n}$. The result is:

$$1 \leq \frac{I(2n)}{I(2n+1)} \leq \frac{I(2n-1)}{I(2n+1)} = \frac{2n+1}{2n}$$

$$\lim_{n\to\infty}\frac{I(2n-1)}{I(2n+1)} = \lim_{n\to\infty}\frac{2n+1}{2n} = \lim_{n\to\infty}\frac{\left(2+\left(\frac{1}{n}\right)\right)}{2} = 1$$

$$and\ also \lim_{n\to\infty} 1 = 1$$

Therefore, by squeeze theorem[8]:

$$\lim_{n\to\infty} \frac{I(2n)}{I(2n+1)} = 1$$

$$=> \frac{\pi}{2} \prod_{k=1}^{\infty} \frac{2k-1}{2k} \cdot \frac{2k+1}{2k} = 1$$

Now we can solve for $\frac{\pi}{2}$ by making it the subject-

$$\lim_{n\to\infty} \frac{I(2n)}{I(2n)+1} = 1$$

$$\frac{\pi}{2} \prod_{k=1}^{\infty} \frac{2k-1}{2k} \cdot \frac{2k+1}{2k} = 1$$

$$\therefore \frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{2k}{2k-1} \cdot \frac{2k}{2k+1} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdots$$ [9]

Therefore, when the formula is entered into the spreadsheet, the series estimates that when $k = 300$, the value of π is approximately 3.13898010388211. This value's deviation from the approximated value till 14 decimal places of is 0.00261254970768.

$$3.14159265358979 - 3.13898010388211 = 0.00261254970768.$$

When compared to the Gregory-Leibniz series, this value is more accurate, but it is still far from the real value till 14 decimal places. This series can obtain the number more quickly and with greater accuracy since it employs multiplication in place of addition.

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8 "Squeeze Theorem Intro (Video) | Khan Academy". *Khan Academy*, 2023, https://www.khanacademy.org/math/ap-calculus-ab/ab-limits-new/ab-1-8/v/squeeze-sandwich-theorem#:~:text=The%20squeeze%20(or%20sandwich)%20theorem,the%20limit%20at%20x%3D0. Accessed 15 Feb 2023.

9 "Wallis Product - Wikipedia". *En.Wikipedia.Org*, 2023, https://en.wikipedia.org/wiki/Wallis_product. Accessed 20 Feb 2023.

Table 2:

Approximation of π in a table using John Wallis Product Formula till 300 terms[10]:

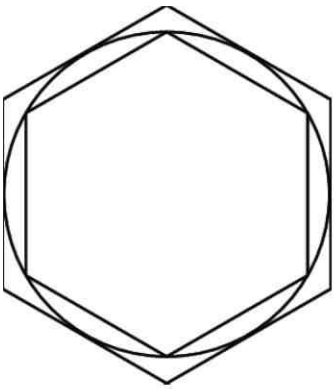
Iteration (number of terms)	Value of π
1	2.66666666666667
2	2.84444444444444
3	2.92571428571429
4	2.97215419501134
5	3.00217595455691
6	3.02317019200136
7	3.03867362888342
8	3.05058999605551
9	3.06003454712689
10	3.06770380664350
…	…
20	3.10351696153923
…	…
188	3.13742885171847
…	…
300	3.12898010388211

[10] Refer to Appendix, John Wallis series data

Archimedes Method-

The first approach I'll look into is estimating the value of utilizing polygons. Archimedes invented this method of discovery. He discovered it by enclosing one regular hexagon inscribed in a circle and another regular hexagon circumscribing the circle. With the help of this procedure, Archimedes was able to determine the exact circumference and diameter of the hexagon and then utilize that information to compute by dividing the circumference by the diameter. He then repeated the procedure, doubling the polygon's sides each time, until it had 96 sides[11].

Fig 1. A showcase of a hexagon is inscribed and circumscribed in a circle[12]:



Since math and technology have advanced significantly since then, allowing us to reduce Archimedes' way of finding, I will be utilizing an approach that is similar to but simpler than the one he employed. This approach keeps the circle's radius constant at 1 unit. I'll first concentrate on calculating the polygon's perimeter inside the circle.

$$S_1 = Side\ BG\ on\ the\ insribed\ hexagon$$

$$\frac{S_1}{2} = Side\ HG\ on\ the\ insribed\ hexagon$$

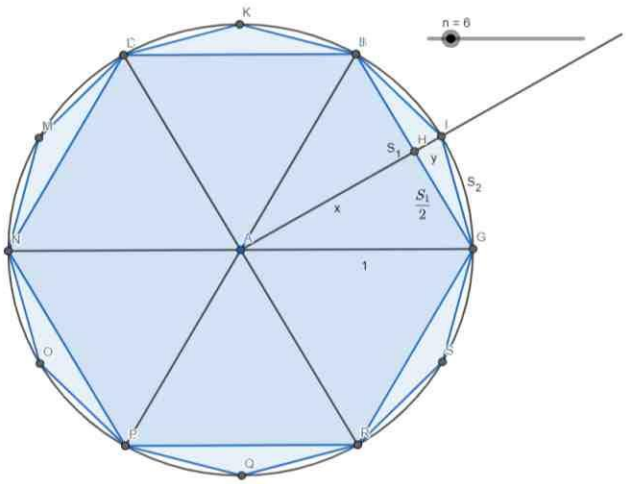
$$S_2 = Arc\ Length$$

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11 "Approximating Pi — NOVA | PBS". *Pbs.Org*, 2023, https://www.pbs.org/wgbh/nova/physics/approximating-pi.html#:~:text=Archimedes'%20method%20finds%20an%20approximation,is%20greater%20than%20the%20circumference). Accessed 15 Feb 2023.

12 *The illustrated method of Archimedes - researchgate.net* (no date). Available at: https://www.researchgate.net/profile/Andre-Assis-7/publication/314487434_The_Illustrated_Method_of_Archimedes_Utilizing_the_Law_of_the_Lever_to_Calculate_Areas_Volumes_and_Centers_of_Gravity/links/58c2bffaaca272e36dd005de/The-Illustrated-Method-of-Archimedes-Utilizing-the-Law-of-the-Lever-to-Calculate-Areas-Volumes-and-Centers-of-Gravity.pdf (Accessed: February 14, 2023).

Fig. 2 – Hexagon and Dodecagon inscribed in a circle[13]-



In the diagram up above, I've built a dodecagon and a hexagon inside of a circle. I concentrated on figuring out how to determine the perimeter of the doubled polygon as Archimedes' method twice the sides of the polygons to be found.

In order to execute this, I first divided the triangle extending from the hexagon's center on one of its sides. I then calculated the distance x between points A and H. I developed the formula for this using the Pythagoras theorem.

From Figure 2 (Using Pythagorean Theorem):

$$x^2 + \left(\frac{S_1}{2}\right)^2 = 1^2$$

$x = \sqrt{1 - \left(\frac{S_1}{2}\right)^2}$, where S_1 is a side of the regular hexagon inscribed in the circle

Both of the points x and y are on the radius since the radius is 1. I then calculated the distance between points H and I.

$$y = 1 - x$$

I used this with the Pythagoras theorem to determine S_2, one of the sides of the dodecagon.

[13] "Calculator Suite - Geogebra". Geogebra.Org, 2022, https://www.geogebra.org/calculator. Accessed 18 June 2021.

From Figure 2 (Using Pythagorean Theorem):

$S_2 = \sqrt{y^2 + \left(\frac{S_1}{2}\right)^2}$ where S_1 is a side of the regular hexagon inscribed in the circle and S_2 is the arc length on the circle.

This will be used to find the perimeter.

The easiest method to get the perimeter of the polygon outside the circle is to multiply it by the ratio $\frac{r}{x}$, which in this case would be equal to $\frac{1}{x}$ because the radius's value is fixed at 1 unit.

Instead of receiving the value for 300 sides, as this method doubles the polygon's number of sides each time, I would have to settle with the value that is the closest to 300 sides, which is 384 sides. In this case, the perimeter divided by diameter for the inside polygon was 3.14155760791186 with an accuracy of 0.00003504567794 and for the outside polygon it was 3.14166274705685 with an accuracy of 0.00007009346706. Although this method provides results that are fairly precise when compared to other ways, it is difficult to use and requires many more steps than the others, making it less effective.

Table 3:

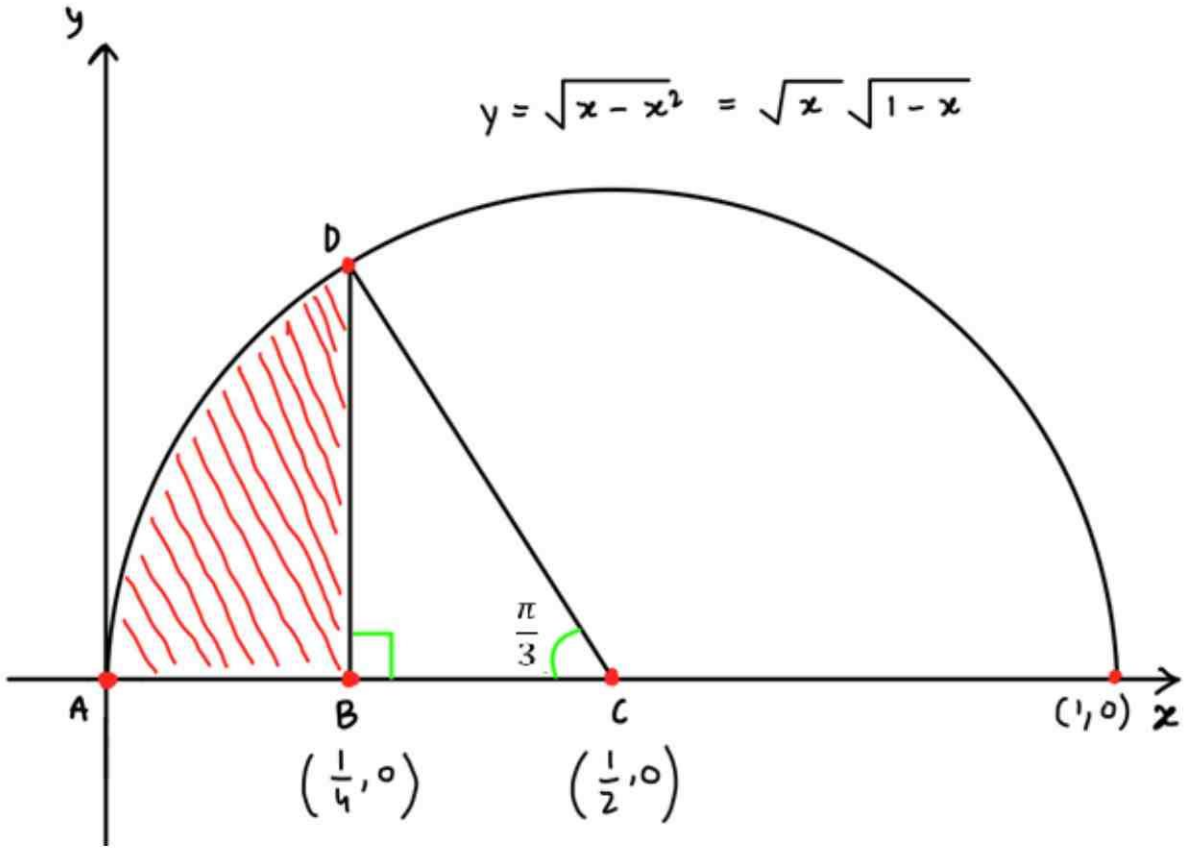
Approximation of π in a table using Archimedes Method[14]:

n	Perimeter (in)	Perimeter (out)	Accuracy(in)	Accuracy(out)
6	3.00000000000000	3.46410161513776	0.14159265358979	0.32250896154796
12	3.10582854123025	3.21539030917347	0.03576411235954	0.07379765558368
24	3.13262861328124	3.15965994209750	0.00896404030856	0.01806728850771
48	3.13935020304687	3.14608621513143	0.00224245054293	0.00449356154164
96	3.14103195089051	3.14271459964537	0.00056070269928	0.00112194605558
192	3.14145247228546	3.14187304997982	0.00014018130433	0.00028039639003
384	3.14155760791186	3.14166274705685	0.00003504567794	0.00007009346706
768	3.14158389214832	3.14161017660469	0.00000876144147	0.00001752301490

[14] Refer to Appendix, Archimedes Method data.

Approximation of π using Newton's Formula-

Fig 3– Semi-circle on a graph with center $(0.5,0)$ and radius 0.5[15]:



Consider the semi-circle with radius 0.5 whose center point is (0.5, 0)

We have equation of the circle radius 0.5 whose center point is (0.5, 0)

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15 "Newton's Formula for Pi." *ProofWiki*, proofwiki.org/wiki/Newton%27s_Formula_for_Pi#:~:text=%CF%80%20(pi)%20can%20be%20approximated,1664%C3%97213%E2%88%92%E2%8B%AF).

We have:

$$\left(x-\tfrac{1}{2}\right)^2+(y-0)^2=\left(\tfrac{1}{2}\right)^2$$

So, equation of above semi-circle radius 0.5 whose center point is (0.5, 0)

⇨ y = $\sqrt{x-x^2}$

⇨ = $\sqrt{x}\sqrt{1-x}$

⇨ = $\sqrt{x}(1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3-\frac{5}{128}x^4-\frac{7}{256}x^5\ldots.)$ (Using general Binomial Theorem when index is a rational number)

We calculate the area in 2 different ways:

We will first calculate the integral between $x=0\ and\ x=\frac{1}{4}$,

$$A=\int_0^{\frac{1}{4}}\left(x^{\frac{1}{2}}-\frac{1}{2}x^{\frac{3}{2}}-\frac{1}{8}x^{\frac{5}{2}}-\frac{1}{16}x^{\frac{7}{2}}-\frac{5}{128}x^{\frac{9}{2}}-\frac{7}{256}x^{\frac{11}{2}}\right)dx$$

$$=\left(\frac{2}{3\times 2^3}-\frac{1}{5\times 2^5}-\frac{1}{28\times 2^7}-\frac{1}{72\times 2^9}-\frac{5}{704\times 2^{11}}-\frac{7}{1664\times 2^{13}}-\cdots\right)\ldots(1)$$

Second, using the techniques of plane geometry we calculate the value of A.

From Fig. 4, in $\Delta\ BCD\ and\ \Delta\ ABD$:

CD = AC

BC = AB

So, by Triangle Identity:

$\Delta\ BCD\cong\Delta\ ABD$

Therefore:

CD = AD = AC, so Δ ACD is equilateral

SO, we know that Δ DBC with angles $30°$, $60°$ and $90°$.

Therefore, using Pythagoras's theorem:

$BD^2 + BC^2 = CD^2$

$BD = (CD^2 - BC^2)^{\frac{1}{2}}$

$= \left(\left(\frac{1}{2}\right)^2 - \left(\frac{1}{4}\right)^2\right)^{\frac{1}{2}}$

$= \frac{1}{4}(4-1)^{\frac{1}{2}}$

$= \frac{\sqrt{3}}{4}$

$Area\ of \diamond ACD = A(s)$

Area of $\Delta\ BCD = A(t)$

$Shaded\ Area = A$

It is observed that

$A =$ Area of the sector ACD ($A(s)$ - $Area$ of the right-angled Δ DCB ($A(t)$.

$A(s)$ is one-sixth of the area of the circle with radius $\frac{1}{2}$

Thus, from Area of the Circle (from Figure 3):

$A(s) = \frac{1}{6}\left(\pi\left(\frac{1}{2}\right)^2\right)$

$A(t)$ is given in terms of side and altitude from Area of triangle (from Figure 3):

$A(t) = \frac{1}{2}\left(\frac{1}{4}\right)\left(\frac{\sqrt{3}}{4}\right)$

Hence:

$A = A(s) - A(t)$

$= \frac{1}{6}\left(\pi\left(\frac{1}{2}\right)^2\right) - \frac{1}{2}\left(\frac{1}{4}\right)\left(\frac{\sqrt{3}}{4}\right)$

$= \frac{\pi}{24} - \frac{\sqrt{3}}{32}$ -------- (2)

Equating equation (1) and equation (2):

$$\frac{\pi}{24} - \frac{\sqrt{3}}{32} = \frac{2}{3\times2^3} - \frac{1}{5\times2^5} - \frac{1}{28\times2^7} - \frac{1}{72\times2^9} - \frac{5}{704\times2^{11}} - \frac{7}{1664\times2^{13}} - \cdots$$

$$\Rightarrow \quad \pi = \frac{3\sqrt{3}}{4} + 21\left(\frac{2}{3\times2^3} - \frac{1}{5\times2^5} - \frac{1}{28\times2^7} - \frac{1}{72\times2^9} - \frac{5}{704\times2^{11}} - \frac{7}{1664\times2^{13}} - \cdots\right)$$

The estimated value of π for $n = 21$ when I created the spreadsheet with this method using the streamlined version was 3.14159265358979. This value is a best match for what it actually is. This value is accurate to the 14th decimal place and efficient because it required the fewest iterations to arrive to the value of π.

Table 4:

Approximation of π in a table using Newton's Formula till 21 terms[16]:

Number of terms	π Value
1	3.00000000000000
2	3.12500000000000
3	3.13906250000000
4	3.14115513392857
5	3.14151117234003
6	3.14157671577487
…	…
21	3.14159265358979

[16] Refer to Appendix, Isaac Newton series data

Gregory-Leibniz Series-

The infinite series produced by Gottfried Leibniz and James Gregory which approximates the value of π:

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots$$

This is one of the earliest series used to precisely approximate the value of π to the 11th decimal point [17]. Utilizing the infinite Taylor series' inverse tan function, James Gregory and Gottfried Leibniz find this series, which was first discovered by the Indian mathematician Madhava in the 1400s.

Proof-

First, we take a sum of an infinite geometric series-

$$S = 1 + a + a^2 + a^3 + a^4 + \cdots$$

In order to express it more succinctly, we can say that as-

$$S \times a = a + a^2 + a^3 + a^4 + \cdots$$

$$S - (S \times a) = 1$$

$$\therefore S = \frac{1}{1-a}$$

And

$$\frac{1}{1-a} = 1 + a + a^2 + a^3 + a^4 + \cdots$$

So, if $a = -u^2$ then

$$\frac{1}{1+u^2} = 1 - u^2 + u^4 - u^6 + u^8 - \cdots$$

Now if we integrate this equation-

$$\int_0^x \frac{1}{1+u^2}\,du = \int_0^x (1 - u^2 + u^4 - u^6 + u^8 - \cdots)\,du$$

$$[arctan(u)]_0^x = [u - \frac{u^3}{3} + \frac{u^5}{5} - \frac{u^7}{7} + \frac{u^9}{9} - \cdots]_0^x$$

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17 "Pi - The Gregory-Leibniz Series". *Crypto.Stanford.Edu*, 2023, https://crypto.stanford.edu/pbc/notes/pi/glseries.html. Accessed 14 Feb 2023.

Integration:

$$arcatn(x) = \left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots\right] - 0$$

$$arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots$$

When putting $x = 1$,

$$\arctan(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots$$

Moreover, we can write this as,

$$\pi = 4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \cdots$$

More simplified version,

$$\pi = 4 \times \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$$

The series calculated an approximation of π as 3.13825932951559 when I applied the formula in the spreadsheet and put $k = 300$. This estimation was slightly larger than the actual value of π by 0.00333332407420, as determined by subtracting the estimation from the true value of π. Despite being close to the starting value, this formula requires more iterations to reach the accurate value of π to 14 decimal places, as opposed to other series which use the sum of all terms. Additionally, it fluctuates greatly because if fewer decimal places are used, the result at certain repeats may be more inaccurate than the one before it. Examples of this include the difference between the values of the 300th and 299th sums. If both values are considered exclusively in terms of two decimal places, the 299th sum's value seems to be more accurate than the 300th sum. As a result, when attempting to determine the approximation of π, this series is ineffective.

Table 5:

Approximation of π in a table using Gregory-Leibniz Series till 300 terms[18]:

S. no.	Temp	Value of $\frac{\pi}{4}$	Value of π	Difference in real value and the found value
1	1	1	4	-0.85840734641021
2	-0.333333333333333	0.666666666666667	2.66666666666667	0.47492598692313
3	0.200000000000000	0.866666666666667	3.46666666666667	-0.32507401307687
4	-0.142857142857143	0.723809523809524	2.89523809523810	0.24635455835170
5	0.111111111111111	0.834920634920635	3.33968253968254	-0.19808988609275
6	-0.090909090909091	0.744011544011544	2.97604617604618	0.16554647754362
7	0.076923076923077	0.820934620934621	3.28373848373848	-0.14214583014869
8	-0.066666666666667	0.754267954267954	3.01707181707182	0.12452083651798
9	0.058823529411765	0.813091483679719	3.25236593471888	-0.11077328112908
10	-0.052631578947368	0.760459904732351	3.04183961892940	0.09975303466039
11	0.047619047619048	0.808078952351398	3.23231580940559	-0.09072315581580
12	-0.043478260869565	0.764600691481833	3.05840276592733	0.08318988766246
13	0.040000000000000	0.804600691481833	3.21840276592733	-0.07681011233754
14	-0.037037037037037	0.767563654444796	3.07025461777919	0.07133803581061
15	0.034482758620690	0.802046413065486	3.20818565226194	-0.06659299867215
16	-0.032258064516129	0.769788348549357	3.07915339419743	0.06243925939237
17	0.030303030303030	0.800091378852387	3.20036551540955	-0.05877286181976
18	-0.028571428571429	0.771519950280959	3.08607980112383	0.05551285246596
19	0.027027027027027	0.798546977307986	3.19418790923194	-0.05259525564215
20	-0.025641025641026	0.772905951666960	3.09162380666784	0.04996884692195
...	...	...	...	...
300	-0.001669449081803	0.784564832378898	3.13825932951559	0.00333332407420

[18] Refer to Appendix, Gregory-Leibniz series data

CONCLUSION-

In my extended essay, the value of π is determined by using and evaluating the Archimedes method, Isaac Newton's Series, Gregory-Leibniz Series, John Wallis Product Formula, and Francois Viète's Series, answering the question, "Which amongst John Wallis Product Formula, Archimedes Method, Francois Viète Formula, Newton's Formula and Gregory-Leibniz Series is the most efficient method to approximate the value of π?"

Under the restrictions, the John Wallis product formula, the Gregory-Leibniz Series, and the Archimedes' method approach failed to approximate the real value of π till 14 decimal places till 300 iterations, however the Francois Viète series succeeded—though more slowly than Isaac Newton's series. Therefore, using Isaac Newton's series would be the most efficient and best technique to determine the value of π. This is due to the fact that, among the formulas chosen, it accurately calculates the value of with the fewest repeats. The reason Newton's formula is the most accurate is that it arrives at the desired number than other formulas that concentrate on calculating the sum or product of a series.

This analysis still has a lot of limitations, though, as I would not be able to easily locate and analyse many formulas because they are not readily available online or in other documents. Additionally, there are algorithms that are more precise and effective, but they require more sophisticated software to run to their full potential and cannot demonstrate their effectiveness on simple software like spreadsheets made in Excel, since that programme only allows for 15 decimal places.

In order to improve in this area, a more thorough investigation can be carried out using Java or Python scripts, which, since the accuracy of the approximation is to more decimal places and also provide the time required to utilise the software, would allow for a more complete examination, aiding in the evaluation of the efficiency of the formula used.

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Appendices-

Appendix 1:

Gregory-Leibniz series-

$$Temp = \frac{(-1)^k}{2k+1}$$

Where $k = S.no. - 1$

S. no.	Temp	Value of $\frac{\pi}{4}$	Value of π	Difference in actual value and calculated value
1	1	1	4	-0.85840734641021
2	-0.333333333333333	0.666666666666667	2.66666666666667	0.47492598692313
3	0.200000000000000	0.866666666666667	3.46666666666667	-0.32507401307687
4	-0.142857142857143	0.723809523809524	2.89523809523810	0.24635455835170
5	0.111111111111111	0.834920634920635	3.33968253968254	-0.19808988609275
6	-0.090909090909091	0.744011544011544	2.97604617604618	0.16554647754362
7	0.076923076923077	0.820934620934621	3.28373848373848	-0.14214583014869
8	-0.066666666666667	0.754267954267954	3.01707181707182	0.12452083651798
9	0.058823529411765	0.813091483679719	3.25236593471888	-0.11077328112908
10	-0.052631578947368	0.760459904732351	3.04183961892940	0.09975303466039
11	0.047619047619048	0.808078952351398	3.23231580940559	-0.09072315581580
12	-0.043478260869565	0.764600691481833	3.05840276592733	0.08318988766246
13	0.040000000000000	0.804600691481833	3.21840276592733	-0.07681011233754
14	-0.037037037037037	0.767563654444796	3.07025461777919	0.07133803581061
15	0.034482758620690	0.802046413065486	3.20818565226194	-0.06659299867215
16	-0.032258064516129	0.769788348549357	3.07915339419743	0.06243925939237
17	0.030303030303030	0.800091378852387	3.20036551540955	-0.05877286181976
18	-0.028571428571429	0.771519950280959	3.08607980112383	0.05551285246596
19	0.027027027027027	0.798546977307986	3.19418790923194	-0.05259525564215
20	-0.025641025641026	0.772905951666960	3.09162380666784	0.04996884692195
21	0.024390243902439	0.797296195569399	3.18918478227760	-0.04759212868780

22	-0.023255813953488	0.774040381615911	3.09616152646364	0.04543112712615
23	0.022222222222222	0.796262603838133	3.18505041535253	-0.04345776176274
24	-0.021276595744681	0.774986008093452	3.09994403237381	0.04164862121599
25	0.020408163265306	0.795394171358758	3.18157668543503	-0.03998403184524
26	-0.019607843137255	0.775786328221503	3.10314531288601	0.03844734070378
27	0.018867924528302	0.794654252749805	3.17861701099922	-0.03702435740943
28	-0.018181818181818	0.776472434567987	3.10588973827195	0.03570291531785
29	0.017543859649123	0.794016294217110	3.17606517686844	-0.03447252327865
30	-0.016949152542373	0.777067141674737	3.10826856669895	0.03332408689085
31	0.016393442622951	0.793460584297688	3.17384233719075	-0.03224968360096
32	-0.015873015873016	0.777587568424672	3.11035027369869	0.03124237989111
33	0.015384615384615	0.792972183809287	3.17188873523715	-0.03029608164736
34	-0.014925373134328	0.778046810674959	3.11218724269983	0.02940541088996
35	0.014492753623188	0.792539564298147	3.17015825719259	-0.02856560360280
36	-0.014084507042254	0.778455057255894	3.11382022902357	0.02777242456622
37	0.013698630136986	0.792153687392880	3.16861474957152	-0.02702209598173
38	-0.013333333333333	0.778820354059547	3.11528141623819	0.02631123735161
39	0.012987012987013	0.791807367046560	3.16722946818624	-0.02563681459645
40	-0.012658227848101	0.779149139198458	3.11659655679383	0.02499609679596
41	0.012345679012346	0.791494818210804	3.16597927284322	-0.02438661925342
42	-0.012048192771084	0.779446625439720	3.11778650175888	0.02380615183091
43	0.011764705882353	0.791211331322072	3.16484532528829	-0.02325267169850
44	-0.011494252873563	0.779717078448509	3.11886831379404	0.02272433979576
45	0.011235955056180	0.790953033504689	3.16381213401876	-0.02221948042896
46	-0.010989010989011	0.779964022515678	3.11985609006271	0.02173656352708
47	0.010752688172043	0.790716710687721	3.16286684275088	-0.02127418916109
48	-0.010526315789474	0.780190394898247	3.12076157959299	0.02083107399680
49	0.010309278350516	0.790499673248763	3.16199869299505	-0.02040603940526
50	-0.010101010101010	0.780398663147753	3.12159465259101	0.01999800099878
51	0.009900990099010	0.790299653246763	3.16119861298705	-0.01960595939726
52	-0.009708737864078	0.780590915382685	3.12236366153074	0.01922899205905

53	0.009523809523810	0.790114724906494	3.16045889962598	-0.01886624603618
54	-0.009345794392523	0.780768930513971	3.12307572205588	0.01851693153391
55	0.009174311926606	0.789943242440577	3.15977296976231	-0.01818031617251
56	-0.009009009009009	0.780934233431568	3.12373693372627	0.01785571986352
57	0.008849557522124	0.789783790953691	3.15913516381477	-0.01754251022497
58	-0.008695652173913	0.781088138779778	3.12435255511911	0.01724009847068
59	0.008547008547009	0.789635147326787	3.15854058930715	-0.01694793571735
60	-0.008403361344538	0.781231785982249	3.12492714392900	0.01666550966080
61	0.008264462809917	0.789496248792167	3.15798499516867	-0.01639234157887
62	-0.008130081300813	0.781366167491353	3.12546466996541	0.01612798362438
63	0.008000000000000	0.789366167491353	3.15746466996541	-0.01587201637562
64	-0.007874015748032	0.781492151743322	3.12596860697329	0.01562404661651
65	0.007751937984496	0.789244089727818	3.15697635891127	-0.01538370532148
66	-0.007633587786260	0.781610501941559	3.12644200776623	0.01515064582356
67	0.007518796992481	0.789129298934040	3.15651719573616	-0.01492454214637
68	-0.007407407407407	0.781721891526632	3.12688756610653	0.01470508748326
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268	-0.001869158878505	0.784465330823446	3.13786132329379	0.00373133029601
269	0.001862197392924	0.786327528216370	3.14531011286548	-0.00371745927569

270	-0.001855287569573	0.784472240646797	3.13788896258719	0.00370369100261
271	0.001848428835490	0.786320669482287	3.14528267792915	-0.00369002433935
272	-0.001841620626151	0.784479048856136	3.13791619542454	0.00367645816525
273	0.001834862385321	0.786313911241457	3.14525564496583	-0.00366299137603
274	-0.001828153564899	0.784485757676557	3.13794303070623	0.00364962288356
275	0.001821493624772	0.786307251301330	3.14522900520532	-0.00363635161553
276	-0.001814882032668	0.784492369268662	3.13796947707465	0.00362317651515
277	0.001808318264014	0.786300687532676	3.14520275013071	-0.00361009654091
278	-0.001801801801802	0.784498885730875	3.13799554292350	0.00359711066629
279	0.001795332136445	0.786294217867320	3.14517687146928	-0.00358421787949
280	-0.001788908765653	0.784505309101667	3.13802123640667	0.00357141718313
281	0.001782531194296	0.786287840295963	3.14515136118385	-0.00355870759406
282	-0.001776198934281	0.784511641361682	3.13804656544673	0.00354608814306
283	0.001769911504425	0.786281552866107	3.14512621146443	-0.00353355787463
284	-0.001763668430335	0.784517884435772	3.13807153774309	0.00352111584671
285	0.001757469244288	0.786275353680060	3.14510141472024	-0.00350876113045
286	-0.001751313485114	0.784524040194946	3.13809616077979	0.00349649281001
287	0.001745200698080	0.786269240893027	3.14507696357211	-0.00348430998231
288	-0.001739130434783	0.784530110458244	3.13812044183298	0.00347221175682
289	0.001733102253033	0.786263212711277	3.14505285084511	-0.00346019725531
290	-0.001727115716753	0.784536096994524	3.13814438797810	0.00344826561170
291	0.001721170395869	0.786257267390393	3.14502906956157	-0.00343641597178
292	-0.001715265866209	0.784542001524184	3.13816800609673	0.00342464749306
293	0.001709401709402	0.786251403233585	3.14500561293434	-0.00341295934455
294	-0.001703577512777	0.784547825720809	3.13819130288323	0.00340135070656
295	0.001697792869270	0.786245618590079	3.14498247436031	-0.00338982077052
296	-0.001692047377327	0.784553571212752	3.13821428485101	0.00337836873878
297	0.001686340640809	0.786239911853561	3.14495964741425	-0.00336699382445
298	-0.001680672268908	0.784559239584654	3.13823695833862	0.00335569525118
299	0.001675041876047	0.786234281460701	3.14493712584280	-0.00334447225301
300	-0.001669449081803	0.784564832378898	3.13825932951559	0.00333332407420

Appendix 2:

John Wallis product formula-

$$A = \frac{2k}{2k-1}$$

$$B = \frac{2k}{2k+1}$$

S.no	A	B	$A \times B$	π	$Actual\ \pi - calculated\ \pi$
1	2.00000000000000	0.66666666666667	1.33333333333333	2.66666666666667	0.47492598692313
2	1.33333333333333	0.80000000000000	1.06666666666667	2.84444444444444	0.29714820914535
3	1.20000000000000	0.85714285714286	1.02857142857143	2.92571428571429	0.21587836787551
4	1.14285714285714	0.88888888888889	1.01587301587302	2.97215419501134	0.16943845857846
5	1.11111111111111	0.90909090909091	1.01010101010101	3.00217595455691	0.13941669903289
6	1.09090909090909	0.92307692307692	1.00699300699301	3.02317019200136	0.11842246158843
7	1.07692307692308	0.93333333333333	1.00512820512821	3.03867362888342	0.10291902470638
8	1.06666666666667	0.94117647058824	1.00392156862745	3.05058999605551	0.09100265753428
9	1.05882352941176	0.94736842105263	1.00309597523220	3.06003454712689	0.08155810646290
10	1.05263157894737	0.95238095238095	1.00250626566416	3.06770380664350	0.07388884694630
11	1.04761904761905	0.95652173913044	1.00207039337474	3.07405516028044	0.06753749330935
12	1.04347826086957	0.96000000000000	1.00173913043478	3.07940134316788	0.06219131042191
13	1.04000000000000	0.96296296296296	1.00148148148148	3.08396341923184	0.05762923435796
14	1.03703703703704	0.96551724137931	1.00127713920817	3.08790206983111	0.05369058375868
15	1.03448275862069	0.96774193548387	1.00111234705228	3.09133688859622	0.05025576499357
16	1.03225806451613	0.96969696969697	1.00097751710655	3.09435872328693	0.04723393030286
17	1.03030303030303	0.97142857142857	1.00086580086580	3.09703782174865	0.04455483184114
18	1.02857142857143	0.97297297297297	1.00077220077220	3.09942935674614	0.04216329684366
19	1.02702702702703	0.97435897435897	1.00069300069300	3.10157726343827	0.04001539015152
20	1.02564102564103	0.97560975609756	1.00062539086929	3.10351696153923	0.03807569205056
21	1.02439024390244	0.97674418604651	1.00056721497448	3.10527732283335	0.03631533075644
22	1.02325581395349	0.97777777777778	1.00051679586563	3.10688211731544	0.03471053627435
23	1.02222222222222	0.97872340425532	1.00047281323877	3.10835109231180	0.03324156127799

24	1.02127659574468	0.97959183673469	1.00043421623969	3.10970078883474	0.03189186475506
25	1.02040816326531	0.98039215686275	1.00040016006403	3.11094516690150	0.03064748668830
26	1.01960784313725	0.98113207547170	1.00036995930448	3.11209609001171	0.02949656357809
27	1.01886792452830	0.98181818181818	1.00034305317324	3.11316370445082	0.02842894913897
28	1.01818181818182	0.98245614035088	1.00031897926635	3.11415673912529	0.02743591446451
29	1.01754385964912	0.98305084745763	1.00029735355338	3.11508274469743	0.02650990889236
30	1.01694915254237	0.98360655737705	1.00027785495971	3.11594828588796	0.02564436770184
31	1.01639344262295	0.98412698412698	1.00026021337497	3.11675909730765	0.02483355628214
32	1.01587301587302	0.98461538461539	1.00024420024420	3.11752021064033	0.02407244294947
33	1.01538461538462	0.98507462686567	1.00022962112514	3.11823605913875	0.02335659445104
34	1.01492537313433	0.98550724637681	1.00021630975557	3.11891056401851	0.02268208957128
35	1.01449275362319	0.98591549295775	1.00020412329047	3.11954720630551	0.02204544728428
36	1.01408450704225	0.98630136986301	1.00019293845263	3.12014908691642	0.02144356667338
37	1.01369863013699	0.98666666666667	1.00018264840183	3.12071897716060	0.02087367642919
38	1.01333333333333	0.98701298701299	1.00017316017316	3.12125936139907	0.02033329219072
39	1.01298701298701	0.98734177215190	1.00016439256946	3.12177247324543	0.01982018034436
40	1.01265822784810	0.98765432098765	1.00015627441788	3.12226032642143	0.01933232716836
41	1.01234567901235	0.98795180722892	1.00014874312063	3.12272474116581	0.01886791242399
42	1.01204819277108	0.98823529411765	1.00014174344437	3.12316736692642	0.01842528666337
43	1.01176470588235	0.98850574712644	1.00013522650439	3.12358970193209	0.01800295165770
44	1.01149425287356	0.98876404494382	1.00012914890869	3.12399311013330	0.01759954345649
45	1.01123595505618	0.98901098901099	1.00012347203358	3.12437883591551	0.01721381767428
46	1.01098901098901	0.98924731182796	1.00011816140848	3.12474801691940	0.01684463667039
47	1.01075268817204	0.98947368421053	1.00011318619128	3.12510169524616	0.01649095834363
48	1.01052631578947	0.98969072164949	1.00010851871948	3.12544082728037	0.01615182630942
49	1.01030927835052	0.98989898989899	1.00010413412475	3.12576629232539	0.01582636126441
50	1.01010101010101	0.99009900990099	1.00010001000100	3.12607890021541	0.01551375337439
51	1.00990099009901	0.99029126213592	1.00009612611747	3.12637939804298	0.01521325554681
52	1.00970873786408	0.99047619047619	1.00009246417013	3.12666847611954	0.01492417747025
53	1.00952380952381	0.99065420560748	1.00008900756564	3.12694677326918	0.01464588032062
54	1.00934579439252	0.99082568807340	1.00008574123296	3.12721488154091	0.01437777204888

55	1.00917431192661	0.99099099099099	1.00008265145880	3.12747335041285	0.01411930317695
56	1.00900900900901	0.99115044247788	1.00007972574344	3.12772269055080	0.01386996303899
57	1.00884955752212	0.99130434782609	1.00007695267411	3.12796337717570	0.01362927641409
58	1.00869565217391	0.99145299145299	1.00007432181345	3.12819585308631	0.01339680050349
59	1.00854700854701	0.99159663865546	1.00007182360124	3.12842053137785	0.01317212221195
60	1.00840336134454	0.99173553719008	1.00006944926731	3.12863779789159	0.01295485569821
61	1.00826446280992	0.99186991869919	1.00006719075455	3.12884801342595	0.01274464016384
62	1.00813008130081	0.99200000000000	1.00006504065041	3.12905151573576	0.01254113785403
63	1.00800000000000	0.99212598425197	1.00006299212598	3.12924862134306	0.01234403224674
64	1.00787401574803	0.99224806201550	1.00006103888177	3.12943962717967	0.01215302641012
65	1.00775193798450	0.99236641221374	1.00005917509912	3.12962481207980	0.01196784150999
66	1.00763358778626	0.99248120300752	1.00005739539689	3.12980443813800	0.01178821545179
67	1.00751879699248	0.99259259259259	1.00005569479254	3.12997875194686	0.01161390164293
68	1.00740740740741	0.99270072992701	1.00005406866721	3.13014798572637	0.01144466786342
69	1.00729927007299	0.99280575539568	1.00005251273434	3.13031235835598	0.01128029523381
70	1.00719424460432	0.99290780141844	1.00005102301138	3.13047207631906	0.01112057727073
71	1.00709219858156	0.99300699300699	1.00004959579428	3.13062733456815	0.01096531902165
72	1.00699300699301	0.99310344827586	1.00004822763443	3.13077831731879	0.01081433627100
73	1.00689655172414	0.99319727891157	1.00004691531785	3.13092519877867	0.01066745481113
74	1.00680272108844	0.99328859060403	1.00004565584623	3.13106814381810	0.01052450977169
75	1.00671140939597	0.99337748344371	1.00004444641984	3.13120730858738	0.01038534500242
76	1.00662251655629	0.99346405228758	1.00004328442194	3.13134284108569	0.01024981250410
77	1.00653594771242	0.99354838709677	1.00004216740460	3.13147488168620	0.01011777190359
78	1.00645161290323	0.99363057324841	1.00004109307582	3.13160356362093	0.00998908996886
79	1.00636942675159	0.99371069182390	1.00004005928775	3.13172901342919	0.00986364016060
80	1.00628930817610	0.99378881987578	1.00003906402594	3.13185135137261	0.00974130221719
81	1.00621118012422	0.99386503067485	1.00003810539954	3.13197069181963	0.00962196177016
82	1.00613496932515	0.99393939393939	1.00003718163227	3.13208714360219	0.00950550998760
83	1.00606060606061	0.99401197604790	1.00003629105426	3.13220081034665	0.00939184324314
84	1.00598802395210	0.99408284023669	1.00003543209439	3.13231179078141	0.00928086280838
85	1.00591715976331	0.99415204678363	1.00003460327347	3.13242017902290	0.00917247456689

86	1.00584795321637	0.99421965317919	1.00003380319778	3.13252606484175	0.00906658874804
87	1.00578034682081	0.99428571428571	1.00003303055326	3.13262953391078	0.00896311967901
88	1.00571428571429	0.99435028248588	1.00003228410008	3.13273066803617	0.00886198555362
89	1.00564971751412	0.99441340782123	1.00003156266768	3.13282954537317	0.00876310821663
90	1.00558659217877	0.99447513812155	1.00003086515016	3.13292624062751	0.00866641296229
91	1.00552486187845	0.99453551912568	1.00003019050207	3.13302082524365	0.00857182834614
92	1.00546448087432	0.99459459459460	1.00002953773446	3.13311336758083	0.00847928600896
93	1.00540540540541	0.99465240641711	1.00002890591126	3.13320393307780	0.00838872051199
94	1.00534759358289	0.99470899470900	1.00002829414594	3.13329258440715	0.00830006918265
95	1.00529100529101	0.99476439790576	1.00002770159838	3.13337938161993	0.00821327196986
96	1.00523560209424	0.99481865284974	1.00002712747199	3.13346438228135	0.00812827130845
97	1.00518134715026	0.99487179487180	1.00002657101103	3.13354764159800	0.00804501199179
98	1.00512820512821	0.99492385786802	1.00002603149811	3.13362921253752	0.00796344105228
99	1.00507614213198	0.99497487437186	1.00002550825192	3.13370914594089	0.00788350764890
100	1.00502512562814	0.99502487562189	1.00002500062502	3.13378749062816	0.00780516296163
101	1.00497512437811	0.99507389162562	1.00002450800186	3.13386429349782	0.00772836009198
102	1.00492610837438	0.99512195121951	1.00002402979695	3.13393959962045	0.00765305396934
103	1.00487804878049	0.99516908212560	1.00002356545305	3.13401345232694	0.00757920126286
104	1.00483091787440	0.99521531100479	1.00002311443959	3.13408589329155	0.00750676029824
105	1.00478468899522	0.99526066350711	1.00002267625116	3.13415696261043	0.00743569097936
106	1.00473933649289	0.99530516431925	1.00002225040607	3.13422669887554	0.00736595471425
107	1.00469483568075	0.99534883720930	1.00002183644503	3.13429513924455	0.00729751434524
108	1.00465116279070	0.99539170506912	1.00002143392991	3.13436231950689	0.00723033408291
109	1.00460829493088	0.99543378995434	1.00002104244261	3.13442827414610	0.00716437944369
110	1.00456621004566	0.99547511312217	1.00002066158392	3.13449303639892	0.00709961719087
111	1.00452488687783	0.99551569506727	1.00002029097255	3.13455663831107	0.00703601527872
112	1.00448430493274	0.99555555555556	1.00001993024415	3.13461911079016	0.00697354279964
113	1.00444444444444	0.99559471365639	1.00001957905042	3.13468048365576	0.00691216993403
114	1.00440528634361	0.99563318777293	1.00001923705827	3.13474078568688	0.00685186790291
115	1.00436681222707	0.99567099567100	1.00001890394903	3.13480004466693	0.00679260892286
116	1.00432900432900	0.99570815450644	1.00001857941772	3.13485828742643	0.00673436616336

117	1.00429184549356	0.99574468085106	1.00001826317231	3.13491553988351	0.00667711370628
118	1.00425531914894	0.99578059071730	1.00001795493312	3.13497182708236	0.00662082650743
119	1.00421940928270	0.99581589958159	1.00001765443215	3.13502717322976	0.00656548036003
120	1.00418410041841	0.99585062240664	1.00001736141252	3.13508160172979	0.00651105186000
121	1.00414937759336	0.99588477366255	1.00001707562796	3.13513513521683	0.00645751837296
122	1.00411522633745	0.99591836734694	1.00001679684219	3.13518779558696	0.00640485800284
123	1.00408163265306	0.99595141700405	1.00001652482855	3.13523960402777	0.00635304956203
124	1.00404858299595	0.99598393574297	1.00001625936946	3.13529058104684	0.00630207254296
125	1.00401606425703	0.99601593625498	1.00001600025600	3.13534074649878	0.00625190709101
126	1.00398406374502	0.99604743083004	1.00001574728753	3.13539011961102	0.00620253397877
127	1.00395256916996	0.99607843137255	1.00001550027125	3.13543871900836	0.00615393458143
128	1.00392156862745	0.99610894941634	1.00001525902190	3.13548656273643	0.00610609085336
129	1.00389105058366	0.99613899613900	1.00001502336133	3.13553366828400	0.00605898530579
130	1.00386100386100	0.99616858237548	1.00001479311824	3.13558005260430	0.00601260098549
131	1.00383141762452	0.99619771863118	1.00001456812785	3.13562573213539	0.00596692145440
132	1.00380228136882	0.99622641509434	1.00001434823158	3.13567072281955	0.00592193077024
133	1.00377358490566	0.99625468164794	1.00001413327680	3.13571504012183	0.00587761346796
134	1.00374531835206	0.99628252788104	1.00001392311655	3.13575869904780	0.00583395454199
135	1.00371747211896	0.99630996309963	1.00001371760929	3.13580171416048	0.00579093942932
136	1.00369003690037	0.99633699633700	1.00001351661868	3.13584409959651	0.00574855399328
137	1.00366300366300	0.99636363636364	1.00001332001332	3.13588586908169	0.00570678450811
138	1.00363636363636	0.99638989169675	1.00001312766656	3.13592703594574	0.00566561764406
139	1.00361010830325	0.99641577060932	1.00001293945628	3.13596761313653	0.00562504045326
140	1.00358422939068	0.99644128113879	1.00001275526474	3.13600761323364	0.00558504035616
141	1.00355871886121	0.99646643109541	1.00001257497831	3.13604704846135	0.00554560512845
142	1.00353356890459	0.99649122807018	1.00001239848738	3.13608593070112	0.00550672288868
143	1.00350877192982	0.99651567944251	1.00001222568617	3.13612427150350	0.00546838208630
144	1.00348432055749	0.99653979238754	1.00001205647252	3.13616208209959	0.00543057149021
145	1.00346020761246	0.99656357388316	1.00001189074781	3.13619937341199	0.00539328017780
146	1.00343642611684	0.99658703071672	1.00001172841678	3.13623615606535	0.00535649752444
147	1.00341296928328	0.99661016949153	1.00001156938740	3.13627244039642	0.00532021319337

148	1.00338983050847	0.99663299663300	1.00001141357074	3.13630823646377	0.00528441712603
149	1.00336700336700	0.99665551839465	1.00001126088083	3.13634355405705	0.00524909953274
150	1.00334448160535	0.99667774086379	1.00001111123457	3.13637840270597	0.00521425088382
151	1.00332225913621	0.99669966996700	1.00001096455160	3.13641279168882	0.00517986190097
152	1.00330033003300	0.99672131147541	1.00001082075421	3.13644673004073	0.00514592354906
153	1.00327868852459	0.99674267100977	1.00001067976718	3.13648022656158	0.00511242702821
154	1.00325732899023	0.99676375404531	1.00001054151777	3.13651328982362	0.00507936376618
155	1.00323624595469	0.99678456591640	1.00001040593555	3.13654592817875	0.00504672541104
156	1.00321543408360	0.99680511182109	1.00001027295234	3.13657814976559	0.00501450382420
157	1.00319488817891	0.99682539682540	1.00001014250216	3.13660996251624	0.00498269107356
158	1.00317460317460	0.99684542586751	1.00001001452106	3.13664137416275	0.00495127942704
159	1.00315457413249	0.99686520376176	1.00000988894712	3.13667239224345	0.00492026134635
160	1.00313479623824	0.99688473520249	1.00000976572037	3.13670302410892	0.00488962948088
161	1.00311526479751	0.99690402476780	1.00000964478265	3.13673327692784	0.00485937666196
162	1.00309597523220	0.99692307692308	1.00000952607764	3.13676315769256	0.00482949589723
163	1.00307692307692	0.99694189602447	1.00000940955069	3.13679267322451	0.00479998036529
164	1.00305810397554	0.99696048632219	1.00000929514886	3.13682183017935	0.00477082341044
165	1.00303951367781	0.99697885196375	1.00000918282078	3.13685063505203	0.00474201853776
166	1.00302114803625	0.99699699699700	1.00000907251663	3.13687909418157	0.00471355940822
167	1.00300300300300	0.99701492537313	1.00000896418807	3.13690721375572	0.00468543983407
168	1.00298507462687	0.99703264094956	1.00000885778821	3.13693499981546	0.00465765377434
169	1.00296735905045	0.99705014749263	1.00000875327154	3.13696245825930	0.00463019533050
170	1.00294985250737	0.99706744868035	1.00000865059386	3.13698959484749	0.00460305874230
171	1.00293255131965	0.99708454810496	1.00000854971230	3.13701641520602	0.00457623838377
172	1.00291545189504	0.99710144927536	1.00000845058520	3.13704292483052	0.00454972875927
173	1.00289855072464	0.99711815561960	1.00000835317212	3.13706912909001	0.00452352449978
174	1.00288184438040	0.99713467048711	1.00000825743375	3.13709503323053	0.00449762035927
175	1.00286532951289	0.99715099715100	1.00000816333195	3.13712064237863	0.00447201121117
176	1.00284900284900	0.99716713881020	1.00000807082960	3.13714596154477	0.00444669204502
177	1.00283286118980	0.99718309859155	1.00000797989068	3.13717099562658	0.00442165796322
178	1.00281690140845	0.99719887955182	1.00000789048014	3.13719574941200	0.00439690417779

179	1.00280112044818	0.99721448467967	1.00000780256392	3.13722022758237	0.00437242600742
180	1.00278551532033	0.99722991689751	1.00000771610892	3.13724443471535	0.00434821887444
181	1.00277008310249	0.99724517906336	1.00000763108293	3.13726837528780	0.00432427830200
182	1.00275482093664	0.99726027397260	1.00000754745462	3.13729205367849	0.00430059991130
183	1.00273972602740	0.99727520435967	1.00000746519354	3.13731547417085	0.00427717941894
184	1.00272479564033	0.99728997289973	1.00000738427003	3.13733864095547	0.00425401263432
185	1.00271002710027	0.99730458221024	1.00000730465526	3.13736155813267	0.00423109545712
186	1.00269541778976	0.99731903485255	1.00000722632115	3.13738422971486	0.00420842387493
187	1.00268096514745	0.99733333333333	1.00000714924039	3.13740665962892	0.00418599396087
188	1.00266666666667	0.99734748010610	1.00000707338638	3.13742885171847	0.00416380187132
189	1.00265251989390	0.99736147757256	1.00000699873323	3.13745080974603	0.00414184384377
190	1.00263852242744	0.99737532808399	1.00000692525572	3.13747253739518	0.00412011619461
191	1.00262467191601	0.99738903394256	1.00000685292928	3.13749403827261	0.00409861531718
192	1.00261096605744	0.99740259740260	1.00000678173002	3.13751531591012	0.00407733767968
193	1.00259740259740	0.99741602067184	1.00000671163462	3.13753637376653	0.00405627982327
194	1.00258397932817	0.99742930591260	1.00000664262038	3.13755721522959	0.00403543836021
195	1.00257069408740	0.99744245524297	1.00000657466518	3.13757784361778	0.00401480997202
196	1.00255754475703	0.99745547073791	1.00000650774747	3.13759826218206	0.00399439140773
197	1.00254452926209	0.99746835443038	1.00000644184623	3.13761847410761	0.00397417948219
198	1.00253164556962	0.99748110831234	1.00000637694098	3.13763848251544	0.00395417107435
199	1.00251889168766	0.99749373433584	1.00000631301175	3.13765829046404	0.00393436312575
200	1.00250626566416	0.99750623441397	1.00000625003906	3.13767790095092	0.00391475263887
201	1.00249376558603	0.99751861042184	1.00000618800394	3.13769731691412	0.00389533667567
202	1.00248138957816	0.99753086419753	1.00000612688785	3.13771654123368	0.00387611235611
203	1.00246913580247	0.99754299754300	1.00000606667273	3.13773557673307	0.00385707685673
204	1.00245700245700	0.99755501222494	1.00000600734097	3.13775442618055	0.00383822740924
205	1.00244498777506	0.99756690997567	1.00000594887537	3.13777309229056	0.00381956129923
206	1.00243309002433	0.99757869249395	1.00000589125914	3.13779157772497	0.00380107586483
207	1.00242130750605	0.99759036144578	1.00000583447592	3.13780988509436	0.00378276849543
208	1.00240963855422	0.99760191846523	1.00000577850972	3.13782801695929	0.00376463663050
209	1.00239808153477	0.99761336515513	1.00000572334495	3.13784597583143	0.00374667775836

210	1.00238663484487	0.99762470308789	1.00000566896638	3.13786376417477	0.00372888941503
211	1.00237529691211	0.99763593380615	1.00000561535913	3.13788138440670	0.00371126918309
212	1.00236406619385	0.99764705882353	1.00000556250869	3.13789883889918	0.00369381469062
213	1.00235294117647	0.99765807962529	1.00000551040088	3.13791612997971	0.00367652361009
214	1.00234192037471	0.99766899766900	1.00000545902185	3.13793325993243	0.00365939365736
215	1.00233100233100	0.99767981438515	1.00000540835808	3.13795023099912	0.00364242259067
216	1.00232018561485	0.99769053117783	1.00000535839634	3.13796704538015	0.00362560820964
217	1.00230946882217	0.99770114942529	1.00000530912373	3.13798370523545	0.00360894835434
218	1.00229885057471	0.99771167048055	1.00000526052763	3.13800021268544	0.00359244090435
219	1.00228832951945	0.99772209567198	1.00000521259572	3.13801656981191	0.00357608377789
220	1.00227790432802	0.99773242630386	1.00000516531594	3.13803277865890	0.00355987493089
221	1.00226757369615	0.99774266365689	1.00000511867651	3.13804884123359	0.00354381235620
222	1.00225733634312	0.99775280898876	1.00000507266594	3.13806475950706	0.00352789408273
223	1.00224719101124	0.99776286353468	1.00000502727296	3.13808053541516	0.00351211817463
224	1.00223713646532	0.99777282850780	1.00000498248656	3.13809617085925	0.00349648273054
225	1.00222717149220	0.99778270509978	1.00000493829599	3.13811166770700	0.00348098588280
226	1.00221729490022	0.99779249448124	1.00000489469073	3.13812702779308	0.00346562579671
227	1.00220750551876	0.99780219780220	1.00000485166048	3.13814225291997	0.00345040066983
228	1.00219780219780	0.99781181619256	1.00000480919518	3.13815734485857	0.00343530873123
229	1.00218818380744	0.99782135076253	1.00000476728498	3.13817230534895	0.00342034824084
230	1.00217864923747	0.99783080260304	1.00000472592025	3.13818713610101	0.00340551748878
231	1.00216919739696	0.99784017278618	1.00000468509157	3.13820183879511	0.00339081479468
232	1.00215982721382	0.99784946236559	1.00000464478971	3.13821641508271	0.00337623850708
233	1.00215053763441	0.99785867237687	1.00000460500564	3.13823086658700	0.00336178700279
234	1.00214132762313	0.99786780383795	1.00000456573054	3.13824519490351	0.00334745868628
235	1.00213219616205	0.99787685774947	1.00000452695576	3.13825940160067	0.00333325198913
236	1.00212314225053	0.99788583509514	1.00000448867283	3.13827348822039	0.00331916536940
237	1.00211416490486	0.99789473684211	1.00000445087348	3.13828745627864	0.00330519731115
238	1.00210526315789	0.99790356394130	1.00000441354960	3.13830130726598	0.00329134632381
239	1.00209643605870	0.99791231732777	1.00000437669323	3.13831504264808	0.00327761094172
240	1.00208768267223	0.99792099792100	1.00000434029662	3.13832866386624	0.00326398972356

241	1.00207900207900	0.99792960662526	1.00000430435213	3.13834217233791	0.00325048125189
242	1.00207039337474	0.99793814432990	1.00000426885232	3.13835556945717	0.00323708413263
243	1.00206185567010	0.99794661190965	1.00000423378988	3.13836885659521	0.00322379699458
244	1.00205338809035	0.99795501022495	1.00000419915765	3.13838203510080	0.00321061848900
245	1.00204498977505	0.99796334012220	1.00000416494863	3.13839510630074	0.00319754728905
246	1.00203665987780	0.99797160243408	1.00000413115594	3.13840807150032	0.00318458208947
247	1.00202839756592	0.99797979797980	1.00000409777286	3.13842093198374	0.00317172160605
248	1.00202020202020	0.99798792756539	1.00000406479280	3.13843368901454	0.00315896457525
249	1.00201207243461	0.99799599198397	1.00000403220929	3.13844634383601	0.00314630975378
250	1.00200400801603	0.99800399201597	1.00000400001600	3.13845889767160	0.00313375591819
251	1.00199600798403	0.99801192842942	1.00000396820673	3.13847135172531	0.00312130186448
252	1.00198807157058	0.99801980198020	1.00000393677539	3.13848370718209	0.00310894640771
253	1.00198019801980	0.99802761341223	1.00000390571602	3.13849596520816	0.00309668838163
254	1.00197238658777	0.99803536345776	1.00000387502277	3.13850812695148	0.00308452663831
255	1.00196463654224	0.99804305283757	1.00000384468991	3.13852019354200	0.00307246004780
256	1.00195694716243	0.99805068226121	1.00000381471182	3.13853216609207	0.00306048749772
257	1.00194931773879	0.99805825242718	1.00000378508299	3.13854404569678	0.00304860789302
258	1.00194174757282	0.99806576402321	1.00000375579801	3.13855583343427	0.00303682015552
259	1.00193423597679	0.99807321772640	1.00000372685159	3.13856753036608	0.00302512322372
260	1.00192678227360	0.99808061420346	1.00000369823853	3.13857913753744	0.00301351605235
261	1.00191938579655	0.99808795411090	1.00000366995372	3.13859065597763	0.00300199761216
262	1.00191204588910	0.99809523809524	1.00000364199217	3.13860208670022	0.00299056688957
263	1.00190476190476	0.99810246679317	1.00000361434897	3.13861343070343	0.00297922288636
264	1.00189753320683	0.99810964083176	1.00000358701929	3.13862468897036	0.00296796461943
265	1.00189035916824	0.99811676082863	1.00000355999843	3.13863586246934	0.00295679112045
266	1.00188323917137	0.99812382739212	1.00000353328175	3.13864695215414	0.00294570143565
267	1.00187617260788	0.99813084112150	1.00000350686469	3.13865795896431	0.00293469462549
268	1.00186915887850	0.99813780260708	1.00000348074279	3.13866888382537	0.00292376976442
269	1.00186219739292	0.99814471243043	1.00000345491168	3.13867972764914	0.00291292594065
270	1.00185528756957	0.99815157116451	1.00000342936704	3.13869049133395	0.00290216225584
271	1.00184842883549	0.99815837937385	1.00000340410467	3.13870117576491	0.00289147782488

272	1.00184162062615	0.99816513761468	1.00000337912042	3.13871178181413	0.00288087177566
273	1.00183486238532	0.99817184643510	1.00000335441021	3.13872231034098	0.00287034324881
274	1.00182815356490	0.99817850637523	1.00000332997006	3.13873276219231	0.00285989139748
275	1.00182149362477	0.99818511796733	1.00000330579605	3.13874313820269	0.00284951538711
276	1.00181488203267	0.99819168173599	1.00000328188433	3.13875343919460	0.00283921439520
277	1.00180831826401	0.99819819819820	1.00000325823111	3.13876366597869	0.00282898761111
278	1.00180180180180	0.99820466786356	1.00000323483268	3.13877381935396	0.00281883423583
279	1.00179533213645	0.99821109123435	1.00000321168540	3.13878390010800	0.00280875348179
280	1.00178890876565	0.99821746880570	1.00000318878568	3.13879390901715	0.00279874457264
281	1.00178253119430	0.99822380106572	1.00000316613001	3.13880384684673	0.00278880674306
282	1.00177619893428	0.99823008849558	1.00000314371493	3.13881371435124	0.00277893923855
283	1.00176991150442	0.99823633156967	1.00000312153704	3.13882351227453	0.00276914131527
284	1.00176366843034	0.99824253075571	1.00000309959302	3.13883324134999	0.00275941223981
285	1.00175746924429	0.99824868651489	1.00000307787959	3.13884290230075	0.00274975128904
286	1.00175131348511	0.99825479930192	1.00000305639352	3.13885249583985	0.00274015774995
287	1.00174520069808	0.99826086956522	1.00000303513165	3.13886202267040	0.00273063091940
288	1.00173913043478	0.99826689774697	1.00000301409087	3.13887148348578	0.00272117010402
289	1.00173310225303	0.99827288428325	1.00000299326814	3.13888087896979	0.00271177462001
290	1.00172711571675	0.99827882960413	1.00000297266044	3.13889020979681	0.00270244379299
291	1.00172117039587	0.99828473413379	1.00000295226483	3.13889947663198	0.00269317695782
292	1.00171526586621	0.99829059829060	1.00000293207840	3.13890868013134	0.00268397345845
293	1.00170940170940	0.99829642248722	1.00000291209831	3.13891782094201	0.00267483264778
294	1.00170357751278	0.99830220713073	1.00000289232175	3.13892689970231	0.00266575388748
295	1.00169779286927	0.99830795262267	1.00000287274597	3.13893591704192	0.00265673654788
296	1.00169204737733	0.99831365935919	1.00000285336826	3.13894487358203	0.00264778000777
297	1.00168634064081	0.99831932773109	1.00000283418595	3.13895376993549	0.00263888365430
298	1.00168067226891	0.99832495812395	1.00000281519643	3.13896260670694	0.00263004688286
299	1.00167504187605	0.99833055091820	1.00000279639712	3.13897138449294	0.00262126909686
300	1.00166944908180	0.99833610648919	1.00000277778549	3.13898010388211	0.00261254970768

Appendix 3:

Francois Viète series-

$$a = c_{S.no.-1}$$

$$b = \sqrt{2}$$

$$c = \frac{b}{2}$$

S.no.	a	b	c	d = Value of π	Actual - d
1	0.00000000000000	1.41421356237310	0.70710678118655	2.82842712474619	0.31316552884360
2	1.41421356237310	1.84775906502257	0.92387953251129	3.06146745892072	0.08012519466907
3	1.84775906502257	1.96157056080646	0.98078528040323	3.12144515225805	0.02014750133174
4	1.96157056080646	1.99036945334439	0.99518472667220	3.13654849054594	0.00504416304385
5	1.99036945334439	1.99759091241034	0.99879545620517	3.14033115695475	0.00126149663504
6	1.99759091241034	1.99939763739241	0.99969881869620	3.14127725093277	0.00031540265702
7	1.99939763739241	1.99984940367829	0.99992470183915	3.14151380114430	0.00007885244549
8	1.99984940367829	1.99996235056520	0.99998117528260	3.14157294036709	0.00001971322270
9	1.99996235056520	1.99999058761915	0.99999529380958	3.14158772527716	0.00000492831263
10	1.99999058761915	1.99999764690340	0.99999882345170	3.14159142151120	0.00000123207859
11	1.99999764690340	1.99999941172576	0.99999970586288	3.14159234557012	0.00000030801968
12	1.99999941172576	1.99999985293144	0.99999992646572	3.14159257658487	0.00000007700492
13	1.99999985293144	1.99999996323286	0.99999998161643	3.14159263433856	0.00000001925123
14	1.99999996323286	1.99999999080821	0.99999999540411	3.14159264877699	0.00000000481281
15	1.99999999080821	1.99999999770205	0.99999999885103	3.14159265238659	0.00000000120320
16	1.99999999770205	1.99999999942551	0.99999999971276	3.14159265328899	0.00000000030080
17	1.99999999942551	1.99999999985638	0.99999999992819	3.14159265351459	0.00000000007520
18	1.99999999985638	1.99999999996409	0.99999999998205	3.14159265357099	0.00000000001880
19	1.99999999996409	1.99999999999102	0.99999999999551	3.14159265358509	0.00000000000470
20	1.99999999999102	1.99999999999776	0.99999999999888	3.14159265358862	0.00000000000117
21	1.99999999999776	1.99999999999944	0.99999999999972	3.14159265358950	0.00000000000029
22	1.99999999999944	1.99999999999986	0.99999999999993	3.14159265358972	0.00000000000007
23	1.99999999999986	1.99999999999996	0.99999999999998	3.14159265358977	0.00000000000002
24	1.99999999999996	1.99999999999999	1.00000000000000	3.14159265358979	0.00000000000000

Appendix 4:

Isaac Newton's formula for π-

$$a = \frac{(2n)!}{2^{4n+1}(n!)^2(2n+1)}$$

$$b = a_{S.no.} + b_{S.no.-1}$$

S.no.	n	a	b	c = Value of π	Actual - c
1	0	0.500000000000000	0.500000000000000	3.000000000000000	0.141592653589793
2	1	0.020833333333333	0.520833333333333	3.125000000000000	0.016592653589793
3	2	0.002343750000000	0.523177083333333	3.139062500000000	0.002530153589793
4	3	0.000348772321429	0.523525855654762	3.141155133928570	0.000437519661221
5	4	0.000059339735243	0.523585195390005	3.141511172340030	0.000081481249763
6	5	0.000010923905806	0.523596119295811	3.141576715774870	0.000015937814926
7	6	0.000002118257376	0.523598237553187	3.141589425319120	0.000003228270671
8	7	0.000000426173210	0.523598663726397	3.141591982358380	0.000000671231410
9	8	0.000000088133247	0.523598751859644	3.141592511157860	0.000000142431930
10	9	0.000000018618793	0.523598770478436	3.141592622870620	0.000000030719175
11	10	0.000000004000824	0.523598774479260	3.141592646875560	0.000000006714231
12	11	0.000000000871721	0.523598775350981	3.141592652105890	0.000000001483905
13	12	0.000000000192142	0.523598775543123	3.141592653258740	0.000000000331054
14	13	0.000000000042767	0.523598775585890	3.141592653515340	0.000000000074455
15	14	0.000000000009599	0.523598775595489	3.141592653572930	0.000000000016862
16	15	0.000000000002170	0.523598775597659	3.141592653585950	0.000000000003842
17	16	0.000000000000494	0.523598775598152	3.141592653588910	0.000000000000879
18	17	0.000000000000113	0.523598775598265	3.141592653589590	0.000000000000202
19	18	0.000000000000026	0.523598775598291	3.141592653589750	0.000000000000046
20	19	0.000000000000006	0.523598775598297	3.141592653589780	0.000000000000010
21	20	0.000000000000001	0.523598775598299	3.141592653589790	0.000000000000000

Appendix 5:

Archimedes Method Data-

n	$\frac{Perimeter}{Diameter}(in)$	$\frac{Perimeter}{Diameter}(out)$	Accuracy(in)	Accuracy(out)
6	3.00000000000000	3.46410161513776	0.14159265358979	0.32250896154796
12	3.10582854123025	3.21539030917347	0.03576411235954	0.07379765558368
24	3.13262861328124	3.15965994209750	0.00896404030856	0.01806728850771
48	3.13935020304687	3.14608621513143	0.00224245054293	0.00449356154164
96	3.14103195089051	3.14271459964537	0.00056070269928	0.00112194605558
192	3.14145247228546	3.14187304997982	0.00014018130433	0.00028039639003
384	3.14155760791186	3.14166274705685	0.00003504567794	0.00007009346706
768	3.14158389214832	3.14161017660469	0.00000876144147	0.00001752301490